

Midterm Exam (Version 1)

CAS CS 132: Geometric Algorithms

Name:

BUID:

- ▷ You will have approximately 75 minutes to complete this exam. Make sure to read every question, some are easier than others.
- ▷ Do not remove any pages from the exam.
- ▷ Your final solution must appear in the solution boxes for each problem. **Only include your final solution in the solution box. You must show your work outside of the solution box.** You will not receive credit if you don't show your work.
- ▷ We will not look at any work on the pages marked “*This page is intentionally left blank.*” You should use these pages for scratch work.

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1 Matrix Equations

$$\begin{bmatrix} 3 & -2 & -6 & 1 \\ 5 & -3 & -3 & 1 \\ 5 & 2 & 1 & 2 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 1 & 1 & 2 & 0 \\ 4 & -1 & 3 & 0 \\ 7 & 1 & 1 & 1 \end{bmatrix} \mathbf{x}$$

(10 points) Determine a general form solution for the above equation. That is, determine a general form solution that describes all vectors in $\mathbb{R}^4$ that satisfy the above equality.

Solution.

x_1

x_2

x_3

x_4

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2 True/False

(8 points) Determine if each of the following statements is **True** or **False**. Bubble in your answers below.

- A. (1 point) For any matrix $A \in \mathbb{R}^{m \times n}$, if $m > n$ then the columns of A are linearly dependent.
- ☐ True
☐ False
- B. (1 point) If A and B are row equivalent and U is an echelon form of A then U is also an echelon form of B .
- ☐ True
☐ False
- C. (1 point) A *consistent* linear system whose coefficient matrix is square and has linearly independent columns has infinitely many solutions.
- ☐ True
☐ False
- D. (1 point) If the augmented matrix of a linear system has a pivot in every column then the linear system is inconsistent.
- ☐ True
☐ False
- E. (1 point) If $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent then so is $\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_3\}$.
- ☐ True
☐ False
- F. (1 point) For $A \in \mathbb{R}^{m \times n}$, if the columns of A span $\mathbb{R}^m$, then the columns of A^T are linearly independent.
- ☐ True
☐ False
- G. (1 point) The matrix that implements a linear transformation is unique.
- ☐ True
☐ False
- H. (1 point) $AB \neq BA$ for all matrices $A, B \in \mathbb{R}^{n \times n}$.
- ☐ True
☐ False

3 Stochastic Matrices

$$A = \begin{bmatrix} 1 & 4 & 1 \\ 2 & 0 & 1 \\ 3 & 5 & 1 \end{bmatrix}$$

- A. (2 points) Calculate $\mathbf{1}^T A$ where $\mathbf{1}$ is the all-ones vector in $\mathbb{R}^3$.

Solution.

- B. (4 points) A matrix B is **stochastic** if its entries are non-negative and $B^T \mathbf{1} = \mathbf{1}$ (this implies the sum of the entries in each column of B is 1). Determine a diagonal matrix C such that AC is stochastic.

Solution.

C. The matrix A is reproduced for convenience.

$$A = \begin{bmatrix} 1 & 4 & 1 \\ 2 & 0 & 1 \\ 3 & 5 & 1 \end{bmatrix}$$

(4 points) Let C be the matrix you determined in the previous part. Calculate $\mathbf{1}^T A C [-1 \ 5 \ 3]^T$.

Solution.

4 Increasing Span

$$[\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3 \ \mathbf{a}_4 \ \mathbf{a}_5 \ \mathbf{a}_6] \sim \begin{bmatrix} 1 & -3 & -11 & -2 & 9 & 1 \\ 0 & -2 & -8 & -4 & 6 & -5 \\ 0 & 0 & 4 & 4 & -2 & 1 \\ 0 & 0 & 0 & 0 & 8 & 5 \end{bmatrix}$$

(10 points) Let $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4, \mathbf{a}_5, \mathbf{a}_6$ be vectors such that the above row equivalence holds. Determine the *first* vector in this list of vectors which lies in the span of the previous vectors in the list. That is, determine the vectors $\mathbf{a}_i$ with the smallest index i such that $\mathbf{a}_i \in \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_{i-1}\}$. Furthermore, write $\mathbf{a}_i$ as a linear combination of the vectors $\mathbf{a}_1, \dots, \mathbf{a}_{i-1}$.

Solution.

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5 Linear Transformations

A.

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} 2x_2 - 3x_1 \\ 0 \\ x_2 + 4x_1 \end{bmatrix}$$

(3 points) Let $S : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be the linear transformation above. Determine the matrix that implements S . That is, determine the matrix A such that $S(\mathbf{v}) = A\mathbf{v}$ for all vectors $\mathbf{v} \in \mathbb{R}^2$.

Solution.

B.

$$x + 3y + z = 0$$

(3 points) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation that projects vectors vertically onto the point-set of the above linear equation. For example,

$$T \left(\begin{bmatrix} 2 \\ -4 \\ 9 \end{bmatrix} \right) = \begin{bmatrix} 2 \\ -4 \\ 10 \end{bmatrix}$$

because $2 + 3(-4) + 10 = 0$. Determine the matrix that implements T .

Solution.

- C. (4 points) Let S and T be the linear transformations from the previous two parts. Determine a **nonzero** vector that is in both the range of S and the range of T (i.e., determine a vector in $\text{ran}(S) \cap \text{ran}(T)$). Justify your answer.

Solution.

Justification.

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